

Mode 10 — Complex

$$\text{C-DP8 } (8 \times 8) \rightarrow \text{DP16 } (8 \times 8) \rightarrow \text{DP16 } (8 \times 4) \rightarrow \text{DP8 } (8 \times 4)$$

Step 1 — C-DP8 (8×8)

$$\text{C-DP8 } (8 \times 8) = D = \sum_{i=0}^7 a_i b_i, \quad a_i = a_i^{re} + j a_i^{im}, \quad b_i = b_i^{re} + j b_i^{im}$$

Step 2 — DP16 (8×8)

$$\text{Re}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re} - \sum_{i=0}^7 a_i^{im} b_i^{im}}_{\text{DP16 } (8 \times 8)}$$

$$\text{Im}(D) = \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im} + \sum_{i=0}^7 a_i^{im} b_i^{re}}_{\text{DP16 } (8 \times 8)}$$

Step 3 — DP16 (8×4) (nibble-split $b = 2^4 b^{hi} + b^{lo}$)

$$\text{Re}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi} - 2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo} - \sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP16 } (8 \times 4)}$$

$$\text{Im}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi} + 2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo} + \sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP16 } (8 \times 4)}$$

Step 4 — DP8 (8×4)

$$\text{Re}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{re,hi}}_{\text{DP8 } (8 \times 4)} - \underbrace{2^4 \sum_{i=0}^7 a_i^{im} b_i^{im,hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{re,lo}}_{\text{DP8 } (8 \times 4)} - \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{im,lo}}_{\text{DP8 } (8 \times 4)}$$

$$\text{Im}(D) = \underbrace{2^4 \sum_{i=0}^7 a_i^{re} b_i^{im,hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{2^4 \sum_{i=0}^7 a_i^{im} b_i^{re,hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{re} b_i^{im,lo}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i^{im} b_i^{re,lo}}_{\text{DP8 } (8 \times 4)}$$